

CLASSROOM CONTACT PROGRAMME

(ACADEMIC SESSION 2019-2020)

PRE-MEDICAL : ENTHUSIAST COURSE [PHASE-II]

TEST # 02

DATE : 22-12-2019

REVISION TEST

Test Pattern : NEET

INSTRUCTIONS

Do not open this Test Booklet until you are asked to do so

1. Immediately fill in the form number on this page of the Test Booklet with Blue/Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The candidates should not write their Form Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **180** questions. The maximum marks are **720**.
5. There are **three** parts in the question paper A,B,C consisting of **Physics, Chemistry** and **Biology** having **45 questions** each in Physics & Chemistry and **90 questions** in Biology part of equal weightage. Each question is allotted 4 (four) marks for **correct** response.
6. **One Fourth** mark will be deducted for indicated incorrect response of each question. **No deduction** from the total score will be made if no response is indicated for an item in the Answer Sheet.
7. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on **Side-1** the Answer Sheet. **Use of pencil is strictly prohibited.**
8. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone any electronic device etc, except the Identity Card inside the examination hall/room.
9. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
10. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Room/Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
11. **Do not fold or make any stray marks on the Answer Sheet.**

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SPACE FOR ROUGH WORK

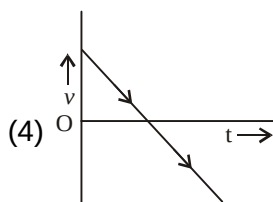
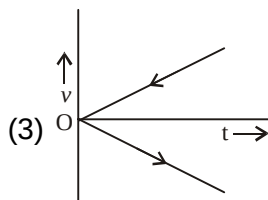
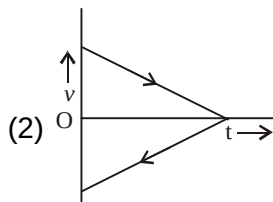
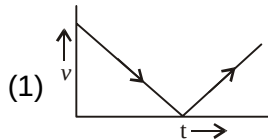
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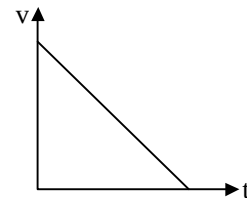
1. A body is projected vertically upwards with a velocity v . It returns to the point from which it was projected after some time. The average velocity and speed for the total time of flight are respectively.

(1) $\frac{v}{2}, v$ (2) $0, \frac{v}{2}$ (3) $0, 0$ (4) $\frac{v}{2}, 0$

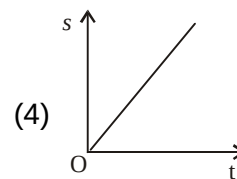
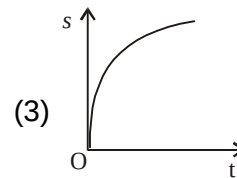
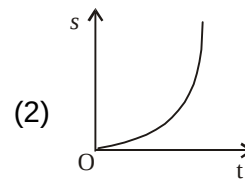
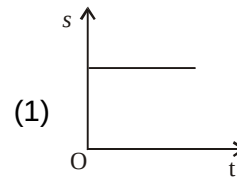
2. A body is thrown vertically upwards with a velocity u . Which of the following graphs represents the variation of velocity with time correctly ?



3. A ball is thrown vertically upwards and its velocity v varies with time t as shown in the figure. Which of the graphs (1), (2), (3), (4) shows the correct



curve of displacement versus time



SPACE FOR ROUGH WORK

4. A particle moving with a constant acceleration from A to B in the straight line AB has velocities u and v at A and B respectively. If C is the mid-point of AB then the velocity of particle while passing C will be
- (1) $\sqrt{\frac{v^2 + u^2}{2}}$ (2) $\frac{v+u}{2}$
- (3) $\frac{v-u}{2}$ (4) $\left(\frac{1}{v} + \frac{1}{u}\right) \frac{1}{2}$
5. An aeroplane is flying in a horizontal direction with a velocity of 900 km/h and at a height of 1960 m. When it is vertically above the point A on the ground, a body is dropped from it. The body strikes the ground at point B . The distance AB will be (take $g = 9.8 \text{ m/s}^2$)
- (1) 3.33 km (2) 5 km
- (3) 10 km (4) 18 km
6. If a body travels half of its total path in the last second of its fall from rest then the total time of its fall will be
- (1) 1.82 s (2) 3.4 s
- (3) 3.41 (4) 3.73
7. A particle moves along the positive branch of the curve $y = x^2/2$ with x governed by $x = t^2/2$ where x and y are measured in metre and t in second. At $t = 2 \text{ s}$, the velocity of the particle is
- (1) $2\mathbf{i} - 4\mathbf{j} \text{ m/s}$ (2) $2\mathbf{i} + 4\mathbf{j} \text{ m/s}$
- (3) $4\mathbf{i} + 2\mathbf{j} \text{ m/s}$ (4) $4\mathbf{i} - 2\mathbf{j} \text{ m/s}$
8. A balloon is ascending at the rate of 12 m/s. When it is at a height of 65 m from the ground, a packet is dropped from it. The packet will reach the ground after time of about (take $g = 10 \text{ m/s}^2$)
- (1) 10 s (2) 2.5 s (3) 5 s (4) 7.5 s
9. The distance x and y along the horizontal plane of a projectile is given by $x = 6t$ and $y = 8t - 5t^2$ in metres where t is in seconds. The angle with the horizontal at which the projectile is projected is
- (1) $\tan^{-1}\left(\frac{3}{4}\right)$ (2) $\tan^{-1}\left(\frac{4}{3}\right)$
- (3) $\sin^{-1}\left(\frac{3}{4}\right)$ (4) $\cos^{-1}\left(\frac{3}{4}\right)$
10. The equation of projectile is $y = \sqrt{3}x - \frac{gx^2}{2}$, The velocity of projection is
- (1) 1.5 m/s (2) 2 m/s
- (3) 2.5 m/s (4) 3 m/s
11. Bullets are fired with velocity u at a fixed angle to the horizontal in all the directions. The maximum area covered by the bullets is
- (1) $\frac{\pi u^2}{g}$ (2) $\frac{\pi^2 u^2}{g}$ (3) $\frac{\pi^2 u^4}{g}$ (4) $\frac{\pi u^4}{g^2}$
12. The angle of projection, at which the horizontal range and maximum height of projectile are equal, is
- (1) 45° (2) $\theta = \tan^{-1}(0.25)$
- (3) $\theta = \tan^{-1} 4$ (4) 60° .

SPACE FOR ROUGH WORK

- | | |
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| <p>13. Three particles, A, B, C are projected from the same point with same initial speeds making angles 30°, 45° and 60° respectively with the horizontal. Which of the following statement is correct ?</p> <p>(1) A, B and C have unequal ranges
 (2) Ranges of A and C are equal and less than that of B
 (3) Ranges of A and C are equal and greater than that of B
 (4) A, B and C have equal ranges.</p> <p>14. A projectile is thrown with an initial velocity of $(a\hat{i} + b\hat{j})$ m/s. If the range of projectile is twice the maximum height reached by it, then</p> <p>(1) $b = a/2$ (2) $b = a$
 (3) $b = 2a$ (4) $b = 4a$</p> <p>15. A projectile is projected with kinetic energy K. If it has the maximum possible horizontal range, then its kinetic energy at the highest point will be</p> <p>(1) $0.25 K$ (2) $0.5 K$
 (3) $0.75 K$ (4) K</p> <p>16. Two seconds after projection, a projectile is moving at 30° above the horizontal; after one more second it is moving horizontally. The initial speed of the projectile is ($g = 10 \text{ m/s}^2$)</p> <p>(1) $10\sqrt{3} \text{ m/s}$ (2) 20 m/s
 (3) 10 m/s (4) $20\sqrt{3} \text{ m/s}$</p> | <p>17. From a point on the ground at a distance of 2 m from the foot of a vertical wall, a ball is thrown at an angle of 45° which just clears the top of the wall and then strikes the ground at a distance of 4 m from the foot of the wall on the other side. The height of the wall is</p> <p>(1) 3 m (2) $\frac{2}{3} \text{ m}$
 (3) $\frac{3}{4} \text{ m}$ (4) $\frac{4}{3} \text{ m}$</p> <p>18. Two stones are projected with the same speed but making different angles with the horizontal. Their ranges are equal. If the angle of projection of one is $\pi/3$ and its maximum height is H, then the maximum height of the other is</p> <p>(1) $3H$ (2) $2H$ (3) $H/3$ (4) $H/2$</p> <p>19. If $\vec{v}_{\text{avg}} = \vec{v}_{\text{instantaneous}}$ for a certain time interval of time, the particle must move :-</p> <p>(1) with variable velocity
 (2) on curved path
 (3) with constant velocity
 (4) with variable acceleration</p> <p>20. When a vector $\vec{A}$ varies with time as $\vec{A} = e^t\hat{i} - t\hat{j} + \sin t\hat{k}$, the angle between the derivative of $\vec{A}$ and the vector $\vec{A}$ at $t = 0$, is :-</p> <p>(1) $\cos^{-1} \frac{2}{\sqrt{3}}$ (2) zero
 (3) $\sin^{-1} \frac{2}{\sqrt{3}}$ (4) $\cos^{-1} \frac{1}{\sqrt{3}}$</p> |
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SPACE FOR ROUGH WORK

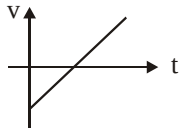
21. The bottom of an electric pole of height 20 m is situated at a distance of $30\sqrt{3}$ m from you due 30° north of east. The magnitude of the position vector of a bulb situated on the top of the pole relative to you is

- (1) 30 m
- (2) 50 m
- (3) $\sqrt{30} \times 10$ m
- (4) $\sqrt{31} \times 10$ m

22. A graph is drawn between square of speed and displacement. Slope of this graph will give :

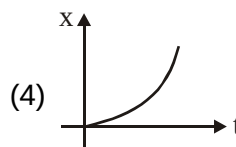
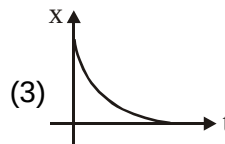
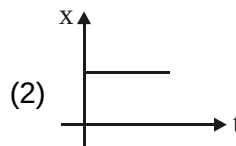
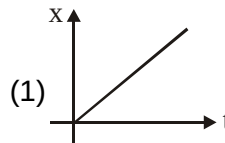
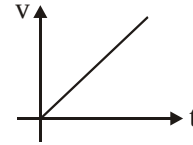
- (1) $\frac{1}{2}$ acceleration
- (2) 2(acceleration)
- (3) displacement
- (4) Time

23. For a moving particle in a straight line the velocity time (v-t) curve is given as shown. Choose correct one :



- (1) The acceleration of particle firstly positive and then negative
- (2) Particle's speed increasing continuously
- (3) Initial velocity of particle is zero
- (4) Particle changes its direction of motion once in a journey.

24. For a rectilinear motion, velocity-time graph is as shown below, which of the following position time graph is correct



25. Choose the INCORRECT alternative.

- (1) It is possible to have a final position of the particle is at origin, while moving with a unidirectional velocity
- (2) For a moving particle distance travelled may decrease
- (3) For a rectilinear motion of a particle if velocity does not change its direction of motion then the average velocity and average speed will be same
- (4) For a rectilinear motion if the direction of acceleration in negative direction speed may be increase.

SPACE FOR ROUGH WORK

26. A ball is dropped from the accelerated plane which is moving in horizontal direction. Choose the correct statement for the ball after releasing from plane. (Air friction is absent)
- Ball will start fall vertically downwards in a straight line.
 - Ball will be accelerated in both vertical and horizontal direction
 - The acceleration of the ball in horizontal direction will be equal to the acceleration of plane
 - Ball will have acceleration in vertical direction only
27. A car is moving with a velocity 54 km/hr and then brakes are applied, which provides a deceleration of 5 m/s^2 . Then the displacement of the car in 10 sec will be :-
- 50 m
 - 40 m
 - 22.5 m
 - 20 m
28. A particle moves with an acceleration $a = k\sqrt{t}$. If initial velocity of particle is zero then velocity of particle as a function of time t is :
- $\frac{2}{3}kt^{3/2}$
 - $\frac{2}{3}kt^{1/3}$
 - $\frac{1}{3}kt^{1/2}$
 - $\frac{2}{3}kt$
29. A particle is projected up. It reaches a particular height at two instants t_1 and t_2 . Then time of flight of particle is
- $\sqrt{t_1 t_2}$
 - $t_1 t_2$
 - $\frac{1}{2}(t_1 + t_2)$
 - $t_1 + t_2$
30. If $\vec{\text{velocity}} \cdot \vec{\text{acceleration}} = 0$ at an instant, then speed of particle at this instant is :
- increasing
 - Decreasing
 - Neither increasing, nor decreasing
 - More information is needed
31. A particle is projected vertically upwards in the presence of uniform gravity ($g=10\text{ m/s}^2$). Its total time of flight is more than 10 second. The distance travelled by particle in last second of its upward journey is :
- 10 m
 - 5 m
 - 2.5 m
 - 1.25 m
32. A particle moves in a straight line so that $s = \sqrt{t}$, then its acceleration is proportional to :
- (Velocity)
 - (Velocity) $^{3/2}$
 - (Velocity) 3
 - (Velocity) 2

SPACE FOR ROUGH WORK

33. A particle moves in a straight line in such a way that its velocity at any point is given by $v^2 = 2 - 3x$, where x is measured from a fixed point. The acceleration is :-

- (1) Uniform
- (2) Zero
- (3) Non-uniform
- (4) indeterminate

34. Ball A is thrown upward from ground at the same time as ball B and with half the speed of ball B. Which of the following is true?

- (1) They will hit the ground at the same time
- (2) A will hit before B
- (3) B will hit before A
- (4) Not enough information is given

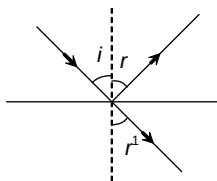
35. Velocity of a particle is given as

$$v = t^2 + 2t - \frac{1}{t}$$

Calculate the net force acting on the particle at time $t = 1$ sec, if the mass of the particle is 10 kg. ($F = ma$)

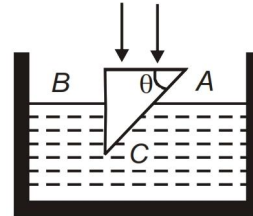
- (1) 15 N (2) 20 N (3) 25 N (4) 50 N

36. A ray of light is incident at an angle i from denser to rare medium. The reflected and the refracted rays are mutually perpendicular. The angle of reflection and the angle of refraction are respectively r and r' , then the critical angle will be



- (1) $\sin^{-1}(\sin r)$ (2) $\sin^{-1}(\tan r')$
- (3) $\sin^{-1}(\tan i)$ (4) $\tan^{-1}(\sin i)$

37. A glass prism of refractive index 1.5 is immersed in water of refractive index $4/3$. A light ray incident normally on face AB is totally reflected at face AC if



- (1) $\sin \theta > 8/9$
- (2) $\sin \theta < 2/3$
- (3) $\sin \theta = \sqrt{3}/2$
- (4) $2/3 \sin \theta \leq 8/9$

38. A concave mirror of focal length 15 cm forms an image having twice the linear dimensions of the object. The position of the object when the image is virtual will be

- (1) 22.5 cm (2) 7.5 cm
- (3) 30 cm (4) 45 cm

39. Two plane mirrors are inclined at an angle of 72° . The number of images of a point object placed symmetrically between them will be

- (1) 2 (2) 3
- (3) 4 (4) 5

40. To get three images of a single object, one should have two plane mirrors at an angle of

- (1) 30° (2) 60°
- (3) 90° (4) 150°

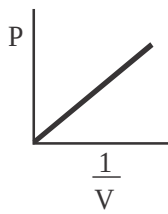
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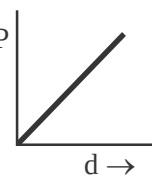
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| <p>41. A man of length h requires a mirror, to see his own complete image of length at least equal to</p> <p>(1) $\frac{h}{4}$</p> <p>(2) $\frac{h}{3}$</p> <p>(3) $\frac{h}{2}$</p> <p>(4) h</p> <p>42. Two plane mirrors are at 45° to each other. If an object is placed between them, then the number of images will be</p> <p>(1) 5</p> <p>(2) 9</p> <p>(3) 7</p> <p>(4) 8</p> <p>43. An object 5 cm tall is placed 1 m from a concave spherical mirror which has a radius of curvature of 20 cm. The size of the image is</p> <p>(1) 0.11 cm</p> <p>(2) 0.50 cm</p> <p>(3) 0.55 cm</p> <p>(4) 0.60 cm</p> | <p>44. The focal length of a concave mirror is 50 cm. Where an object be placed so that its image is two times and inverted</p> <p>(1) 75 cm</p> <p>(2) 72 cm</p> <p>(3) 63 cm</p> <p>(4) 50 cm</p> <p>45. Which one of the following statements is true</p> <p>(1) An object situated at the principle focus of a concave lens will have its image formed at infinity</p> <p>(2) Concave mirror can give diminished virtual image</p> <p>(3) Given a point source of light, a convex mirror can produce a parallel beam of light</p> <p>(4) The virtual image formed in a plane mirror can be photographed</p> |
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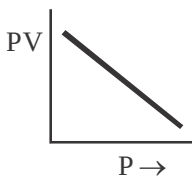
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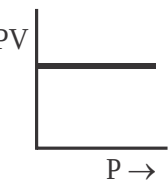
46. The uncertainty in position of an electron & helium atom are same. If the uncertainty in momentum for the electron is 32×10^5 , then the uncertainty in momentum of helium atom will be
 (1) 32×10^5 (2) 16×10^5
 (3) 8×10^5 (4) None
47. In an atom, for how many electrons, the quantum numbers will be, $n = 3$, $\ell = 2$, $m = +2$, $s = +\frac{1}{2}$:-
 (1) 18 (2) 6
 (3) 24 (4) 1
48. For the azimuthal quantum number (ℓ), the total number of magnetic quantum number is given by:-
 (1) $\ell = \frac{(m+1)}{2}$ (2) $\ell = \frac{(m-1)}{2}$
 (3) $\ell = \frac{(2m+1)}{2}$ (4) $\ell = \frac{(2m-1)}{2}$
49. An atom has 2 electrons in K-shell, 8 electrons in L-shell & 8 electrons in M-shell. The number of s-electrons presents in the element is :-
 (1) 10 (2) 7 (3) 6 (4) 4
50. Bohr's theory is not applicable to -
 (1) He (2) Li^{2+}
 (3) He^{2+} (4) Both (1) & (3)
51. If velocity of an electron in 1st orbit of H atom is V, what will be the velocity of 3rd orbit of Li^{+2} -
 (1) V (2) V/3 (3) 3 V (4) 9 V
52. Which electronic level would allow the hydrogen atom to absorb a photon but not to emit a photon :-
 (1) 3s (2) 2p
 (3) 2s (4) 1s
53. The third line in Balmer series corresponds to an electronic transition between which Bohr's orbits in hydrogen :-
 (1) $5 \rightarrow 3$ (2) $5 \rightarrow 2$
 (3) $4 \rightarrow 3$ (4) $4 \rightarrow 2$
54. If λ_0 is the threshold wavelength for photoelectric emission, λ wavelength of light falling on the surface of metal, and m mass of electron, then de Broglie wavelength of emitted electron is :-
 (1) $\left[\frac{h(\lambda\lambda_0)}{2mc(\lambda_0 - \lambda)} \right]^{\frac{1}{2}}$ (2) $\left[\frac{h(\lambda_0 - \lambda)}{2mc\lambda\lambda_0} \right]^{\frac{1}{2}}$
 (3) $\left[\frac{h(\lambda - \lambda_0)}{2mc\lambda\lambda_0} \right]^{\frac{1}{2}}$ (4) $\left[\frac{h\lambda\lambda_0}{2mc} \right]^{\frac{1}{2}}$
55. The shortest wavelength of He^+ in Balmer series is x, then longest wavelength in the Paschene series of Li^{+2} is :-
 (1) $\frac{36x}{5}$ (2) $\frac{16x}{7}$
 (3) $\frac{9x}{5}$ (4) $\frac{5x}{9}$

SPACE FOR ROUGH WORK

56. An open vessel at 27°C is heated until $3/8$ th of the air in it has been expelled. Assuming that the volume of vessel remains constant, calculate the temperature at which the vessel was heated.
- (1) 307°C (2) 107°C
(3) 480°C (4) 207°C
57. A bottle of dry ammonia and one of dry hydrogen chloride are connected through a long tube. The stoppers at both ends of the tube are opened simultaneously. The white ammonium chloride ring first formed will be
- (1) at the centre of the tube
(2) near the hydrogen chloride bottle
(3) near the ammonia bottle
(4) throughout the length of the tube
58. To which of the following gaseous mixtures is Dalton's law not applicable?
- (1) $\text{Ne} + \text{He} + \text{SO}_2$
(2) $\text{NH}_3 + \text{HCl} + \text{HBr}$
(3) $\text{O}_2 + \text{N}_2 + \text{CO}_2$
(4) $\text{N}_2 + \text{H}_2 + \text{O}_2$
59. A bubble of gas released at the bottom of a lake increases to eight times its original volume when it reaches the surface. Assuming that one atm pressure is equivalent to the pressure exerted by a column of water 10 m high, what is the depth of the lake?
- (1) 90 m (2) 10 m
(3) 70 m (4) 80 m
60. At what temperature is the rms speed of hydrogen molecules the same as that of oxygen molecules at 1327°C ?
- (1) 173 K (2) 100 K
(3) 400 K (4) 523 K
61. At low pressure, the van der Waals equation is reduced to
- (1) $Z = \frac{pV_m}{RT} = 1 - \frac{a}{RTV_m}$
(2) $Z = \frac{pV_m}{RT} = 1 + \frac{b}{RT}p$
(3) $pV_m = RT$
(4) $Z = \frac{pV_m}{RT} = 1 - \frac{a}{RT}$
62. The temperature at which real gases obey the ideal gas laws over a wide range of pressures is called
- (1) critical temperature
(2) inversion temperature
(3) Boyle temperature
(4) reduced temperature
63. Which is not a correct representation of Boyle's law?
- (1) 

(2) 

(3) 

(4) 

SPACE FOR ROUGH WORK

64. For gaseous state, if most probable speed is denoted by C^* , average speed by $\bar{C}$ and root mean square speed by C , then for a large number of molecules, the ratios of these speeds are

(1) $C^* : \bar{C} : C = 1.225 : 1.128 : 1$

(2) $C^* : \bar{C} : C = 1.228 : 1.225 : 1$

(3) $C^* : \bar{C} : C = 1 : 1.128 : 1.225$

(4) $C^* : \bar{C} : C = 1 : 1.225 : 1.128$

65. The ratio of van der Waals' constants a and b , $\left(\frac{a}{b}\right)$ has the dimension of

(1) atm L^{-1} (2) L atm mol^{-1}

(3) L mol^{-1} (4) atm L mol^{-2}

66. An isomer of ethanol is :

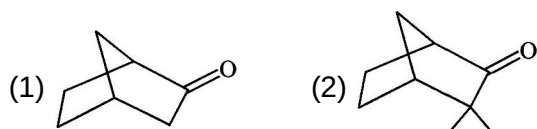
(1) methanol (2) diethyl ether

(3) acetone (4) dimethyl ether

67. Total number of position isomers of dimethyl cyclohexane.

(1) 2 (2) 3 (3) 4 (4) 5

68. Among the following compounds, one that will not show keto-enol tautomerism is



69. The Z-configuration in the following is



70. An alkene exists as geometrical isomers. The minimum number of carbon atoms in the molecules is

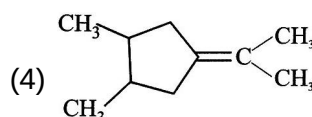
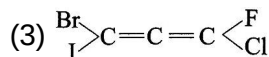
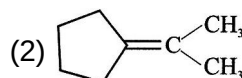
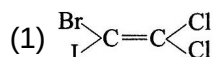
(1) 1 (2) 2

(3) 3 (4) 4

71. The number of geometrical isomers for $\text{CH}_3 - \text{CH} = \text{CH} - \text{CH} = \text{CH} - \text{CH} = \text{CH}_2$ is

(1) 2 (2) 4 (3) 6 (4) 8

72. Which of the following compound can show geometrical isomerism?



73. The smallest alcohol which exhibit optical activity is

(1) tert-Butyl alcohol

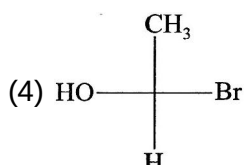
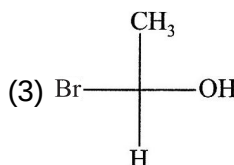
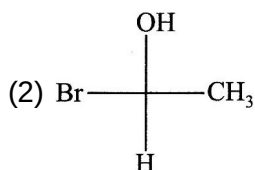
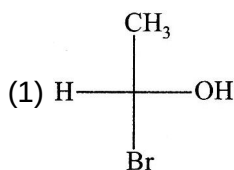
(2) Butan-1-ol

(3) Pentan-2-ol

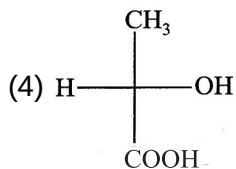
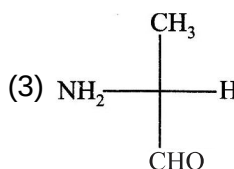
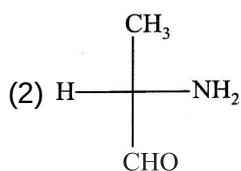
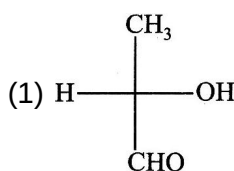
(4) Butan-2-ol

SPACE FOR ROUGH WORK

74. Which of the following compound has 'S' configuration?



75. Which of the following compound has 'D' configuration?



76. An alloy of copper, silver and gold is found to have copper constituting the C.C.P. lattice. If silver atoms occupy the edge centres and gold is present at body centre, the alloy has a formula-

- (1) $\text{Cu}_4\text{Ag}_2\text{Au}$ (2) $\text{Cu}_4\text{Ag}_4\text{Au}$
(3) $\text{Cu}_4\text{Ag}_3\text{Au}$ (4) CuAgAu

77. Which is/are amorphous solids-

- (1) Rubber (2) Plastics
(3) Glass (4) All

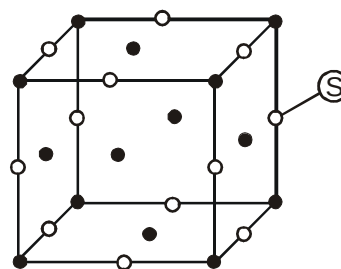
78. The vacant space in B.C.C. unit cell is-

- (1) 32% (2) 10%
(3) 23% (4) 46%

79. In a solid, oxide ions are arranged in CCP. Cations A occupy one-sixth of the tetrahedral voids and cations B occupy one-third of the octahedral voids. The formula of the compound is -

- (1) ABO_3 (2) AB_2O_3
(3) A_2BO_3 (4) $\text{A}_2\text{B}_2\text{O}_3$

80. For the structure given below the site marked as S is a-



- (1) Tetrahedral void
(2) Cubic void
(3) Octahedral void
(4) None of these

81. Schottky defect is noticed in-

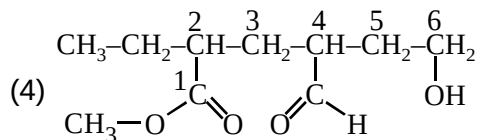
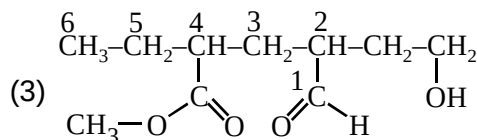
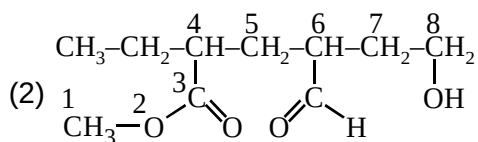
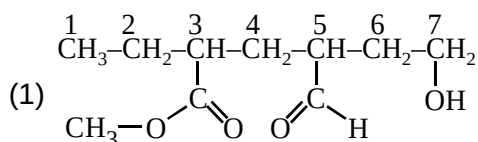
- (1) NaCl (2) KCl
(3) CsCl (4) All

82. A cubic crystal possesses total elements of symmetry-

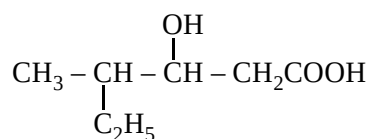
- (1) 9 (2) 13
(3) 1 (4) 23

SPACE FOR ROUGH WORK

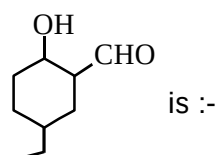
83. The co-ordination number of a body centred atom is-
(1) 4 (2) 6 (3) 8 (4) 12
84. The decreasing order of the size of void is
(1) Cubic > Octahedral > Tetrahedral > Trigonal
(2) Trigonal > Tetrahedral > Octahedral > Cubic
(3) Trigonal > Octahedral > Tetrahedral > Cubic
(4) Cubic > Tetrahedral > Octahedral > Trigonal
85. In $A_x B_y$ atoms of A occupy each corner of the cube and atom of B occupy centre of each face. The composition will be -
(1) AB_3 (2) $A_2 B_3$ (3) $A_3 B$ (4) $A_3 B_4$
86. In IUPAC nomenclature system correct priority order of functional group in organic compounds are :-
(1) $-\text{CONH}_2$, $-\text{COOR}$, $>\text{CO}$, OH
(2) $-\text{COOR}$, $-\text{CONH}_2$, OH, $>\text{CO}$
(3) $-\text{CONH}_2$, $>\text{CO}$, CONH_2 , OH
(4) $-\text{COOR}$, $-\text{CONH}_2$, $>\text{CO}$, OH
87. Which of the following represent correct numbering of parent chain according to IUPAC system :-



88. IUPAC name of given compound is :-



- (1) 4-Ethyl-3-hydroxypentanoic acid.
(2) 3-Hydroxy-4-ethylpentanoic acid
(3) 3-Hydroxy-4-methylhexanoic acid
(4) 4-Ethyl-3-hydroxy-4-methylbutanoic acid
89. The correct IUPAC name of 2-cyano-4-formyl- 5-methylhexane :-
(1) 5-Methyl-2-cyano-4-formylhexane
(2) 4-Formyl-3,5-dimethylpentanecyanide
(3) 4-Formyl-2,5-dimethylhexanenitrile
(4) 4-formyl-2,5-dimethylhexanecyanide
90. The IUPAC name of the compound



- (1) 3-Ethyl-6-hydroxycyclohexanecarbaldehyde
(2) 5-Ethyl-2-hydroxycyclohexanecarbaldehyde
(3) 2-Formyl-4-ethylcyclohexanol
(4) 1-Formyl-5-ethyl-2-hydroxycyclohexane

SPACE FOR ROUGH WORK

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|---|---|
| <p>91. In food chain, the total amount of living material, is shown by :-</p> <ol style="list-style-type: none"> (1) Pyramid of energy (2) Pyramid of numbers (3) Pyramid of biomass (4) All of these <p>92. What is correct for decomposers :-</p> <ol style="list-style-type: none"> (1) They convert inorganic material to organic material (2) They play significant role in mineral cycle (3) They convert solar energy into chemical energy (4) They decrease inorganic material in ecosystem <p>93. Father of ecosystem ecology is :-</p> <ol style="list-style-type: none"> (1) Hackel (2) Reiter (3) Prof RAM Deo Mishra (4) E.P. Odum <p>94. What is true for ecotone :-</p> <ol style="list-style-type: none"> (1) It is transitional zone between two different species (2) It is transitional zone between two different population (3) It is transitional zone between different community (4) Minimum species diversity in ecotone <p>95. In any Grassland and forest community edge species are :-</p> <ol style="list-style-type: none"> (1) Those species which occurs in ecotone (2) Those species which increased their variety and density at community border (3) Those species which spend the greatest amount of their time in junction of forest and grass land (4) All of the above | <p>96. If Biotic potential of a species is high then :-</p> <ol style="list-style-type: none"> (1) No. of species in area will increase (2) Species diversity will increase in area (3) No. of individuals of particular species will increase (4) No. of individuals of other species will decrease <p>97. <i>Cuscutta</i> and <i>Rafflesia</i> are :-</p> <ol style="list-style-type: none"> (1) Producer (2) Primary consumer (3) Secondary consumer (4) Both (1) and (2) <p>98. What type of ecological succession would operate after a volcanic eruption and forest fire :-</p> <ol style="list-style-type: none"> (1) Primary (2) Secondary (3) Primary & secondary respectively (4) Secondary and primary respectively <p>99. Allelopathy refers to :-</p> <ol style="list-style-type: none"> (1) System of medicine based on plant products (2) A disease caused by a pathogen (3) Study of chemical interaction between two plant species (4) Competition between two plant species <p>100. <i>Penicillium</i> does not allow the growth of bacterium <i>Staphylococcus</i>. This type of relationship is called :-</p> <ol style="list-style-type: none"> (1) Commensalism (2) Antibiosis (3) Allelopathy (4) Mutualism <p>101. Helotism found in :-</p> <ol style="list-style-type: none"> (1) Epiphytes (2) Termites and Flagellates (3) Lichen (4) Birds and tree |
|---|---|

102. Which zone of lake has minimum species diversity :-
 (1) Littoral zone
 (2) Limnetic zone
 (3) Profundal zone
 (4) All of the above
103. Clear stratification is found in :-
 (1) Tropical rain forest
 (2) Deciduous forest
 (3) Temperate evergreen forest
 (4) Alpine forest
104. Limnetic zone is :-
 (1) The shallow water zone around the edge of lake which supports rooted vegetation
 (2) The open water zone beyond the littoral zone
 (3) The zone where light does not reach
 (4) Both (1) and (2)
105. Pioneer community of lithosere is :-
 (1) Crustose moss
 (2) Foliose lichens stage
 (3) Plankton stage
 (4) Crustose lichens stage
106. Competition is the most severe between :-
 (1) Closely related species growing in different niche
 (2) Closely related species growing in the same habitat
 (3) Distantly related species growing in the same habitat
 (4) Distantly related species growing in different niches
107. What is not true for primary succession.
 (1) Humus or organic matter is absent in the early stages
 (2) The area is barren from the beginning
 (3) Reproductive structures of previous living beings may be present.
 (4) Takes more time 1000 year

108. A forest community has been destroyed due to fire, if new community again develops on this area, then it is :-
 (1) Primary succession
 (2) Secondary succession
 (3) Both
 (4) All of the above
109. Retrogressive succession is :-
 (1) Forest converts into grass land
 (2) Desert converts into grass land
 (3) Grassland converts into forest
 (4) Both (1) and (3)
110. The storage of energy at consumer level is known as :-
 (1) Net primary productivity
 (2) Net productivity
 (3) Gross primary production
 (4) Secondary productivity
111. What is true for following food web :-
-
- ```

graph BT
 A --> E
 B --> E
 C --> G
 D --> H
 E --> I
 G --> J
 H --> J
 I --> K
 J --> K

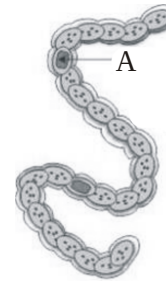
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- (1) K is decomposer  
 (2) B is herbivorous  
 (3) H is secondary consumer  
 (4) J is predator
112. Destruction of stratospheric ozone is mainly due to :-  
 (1) Active nitrogen  
 (2) Active hydrogen  
 (3) Active oxygen  
 (4) Active chlorine
113. Inverted pyramid is found in :-  
 (1) Biomass pyramid of aquatic system  
 (2) Energy pyramid of grass land  
 (3) Biomass pyramid of grass land  
 (4) Pyramid of number of aquatic system

114. Mr. Bean is eating curd / yoghurt. For this food intake in a food chain he should be considered as occupying
- First trophic level
  - Second trophic level
  - Third trophic level
  - Fourth trophic level
115. Dispersal of spores of slime moulds occur through :-
- Air
  - Water
  - Insects
  - Crustaceans
116. Identify the incorrect information about given fungi :-
- Puccinia – It causes wheat rust
  - Neurospora – Extensively used in genetics
  - Claviceps – Source of LSD
  - Ustilago – It is known as rust fungi
117. Deuteromycetes is also known as fungi imperfecti because :-
- Members are not well differentiated
  - Mycellium is aseptate and coenocytic
  - Members do not produce zoospores
  - It lacks sexual reproduction
118. Presence of some bacteria is expected in hot springs. These are :-
- Archaeobacteria
  - Phototrophic autotrophs
  - Chemosynthetic autotrophs
  - Heterotrophic eubacteria
119. Identify the correct statement :-
- All plants without any exception are obligate autotrophs
  - All fungi without any exception are obligate saprotrophs
  - All animals without any exception are obligate heterotrophs
  - All monerans without any exception are obligate anaerobes

120. Identify the incorrect match :-

| Class of fungi     | Sexual/Asexual spores | Example    |
|--------------------|-----------------------|------------|
| (1) Phycomycetes   | Oospores              | Rhizopus   |
| (2) Ascomycetes    | Ascospores            | Neurospora |
| (3) Basidiomycetes | Conidia               | Agaricus   |
| (4) Deuteromycetes | Conidia               | Alternaria |

121. Diagram of Nostoc is given. Select the correct statement :-

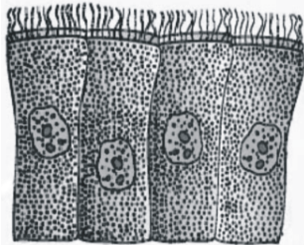


- A is mucilaginous sheath and is also found in Rivularia
  - A is mucilaginous sheath and is also found in Anabaena
  - A is heterocyst and is also found in methanogens
  - A is heterocyst and is also found in nostoc
122. Identify the incorrect statement :-
- Conidia in ascomycetes are exogenously produced
  - Basidiospores in mushrooms are exogenously produced
  - Zoospores in Mucor are endogenously produced
  - Ascospores in sac fungi are exogenously produced
123. Select the option with edible fungi :-
- Buffles only
  - Truffles only
  - Both buffles and morels
  - Both morels and truffles
124. Most of the bacteria obtain nutrition from
- Sunlight
  - Inorganic substrates
  - Dead or decaying matter
  - Plants

125. Which of the following is not applicable to zoological park?
- (1) In vivo mode of conservation
  - (2) Wild animals are kept under human care
  - (3) Wild animals are kept in separate enclosures
  - (4) All are true
126. In a biological name - *Mangifera indica* Linn., Linn indicates :-
- (1) Subspecies
  - (2) Breed
  - (3) Variety
  - (4) Author
127. Key is used as a taxonomic aid for identification of plants and animals. What is correct for the key?
- (1) Couplet is the pair of contrasting characters
  - (2) Keys can be used for different taxons
  - (3) Each statement in the key is termed as lead
  - (4) All of these
128. The main difference between gram positive and gram negative bacteria in the composition of
- (1) Cillia
  - (2) Cell wall
  - (3) Cell membrane
  - (4) Cytoplasm
129. Mycoplasma are smallest living cell known and can survive without
- (1) Cell membrane
  - (2) Oxygen
  - (3) Heterotrophic nutrition
  - (4) Genetic material
130. *Rhodospirillum* is a
- (1) Purple sulphur bacteria
  - (2) Purple nonsulphur bacteria
  - (3) Green sulphur bacteria
  - (4) Green nonsulphur bacteria
131. Characteristic photosynthetic pigments present in BGA are
- (1) Chlorophyll a, Chlorophyll b
  - (2) Chlorophyll a, phycocyanin and phycoerythrin
  - (3) Chlorophyll c, Phycobilins
  - (4) Chlorophyll c, phycocyanin and phycoerythrin
132. Which of the following played a significant role in the evolution of O<sub>2</sub> on earth?
- (1) Archaeobacteria
  - (2) BGA
  - (3) Green Algae
  - (4) Protistan algae
133. Algal bloom in the polluted water bodies are often formed by
- (1) Blue green algae
  - (2) Sulphur bacteria
  - (3) Nitrifying bacteria
  - (4) Archaeobacteria
134. A longitudinal and a transverse flagellum are feature of organisms
- (1) Diatoms
  - (2) Sporozoans
  - (3) Slime moulds
  - (4) Dinoflagellates
135. Which organisms undergo such a rapid multiplication that they make the sea appear red (called red tides)?
- (1) *Navicula*
  - (2) *Trichodesmium*
  - (3) *Gonyaulax*
  - (4) *Chlamydomonas*

- |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
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| <p>136. Simple epithelium is made of :-<br/>         (1) Actively dividing cells<br/>         (2) Several layers of cells<br/>         (3) Compactly packed single layer of cells<br/>         (4) Non-cellular layer of hyaluronic acid</p> <p>137. Goblet cells are :-<br/>         (1) Multicellular glands<br/>         (2) Stratified epithelium<br/>         (3) Glial cells<br/>         (4) Unicellular glands</p> <p>138. Sebaceous glands are :-<br/>         (1) Holocrine (2) Eccrine<br/>         (3) Merocrine (4) Epicrine</p> <p>139. Inner lining of gut and liver is made of :-<br/>         (1) Simple columnar epithelium<br/>         (2) Simple cuboidal epithelium<br/>         (3) Simple squamous epithelium<br/>         (4) Pseudostratified epithelium</p> <p>140. Stratified and non-keratinized squamous epithelium occurs in :-<br/>         (1) Vagina (2) Buccal cavity<br/>         (3) Stomach (4) Both (1) and (2)</p> <p>141. Epithelium forming the inner layer of urinary bladder is :-<br/>         (1) Transitional<br/>         (2) Myoepithelium<br/>         (3) Neurosensory<br/>         (4) Simple squamous</p> <p>142. Pseudostratified columnar epithelium is found in :-<br/>         (1) Stomach (2) Vagina<br/>         (3) Trachea (4) Mouth</p> <p>143. Epithelium present in cornea, oesophagus and vagina is :-<br/>         (1) Stratified columnar<br/>         (2) Stratified squamous<br/>         (3) Ciliated<br/>         (4) Pseudostratified</p> <p>144. Ciliated epithelium lines :-<br/>         (1) Trachea (2) Oesophagus<br/>         (3) Stomach (4) Duodenum</p> <p>145. Tessellated epithelium is found in :-<br/>         (1) Lining of blood vessel<br/>         (2) Lining of intestine<br/>         (3) Seminiferous tubule<br/>         (4) Tubular parts of nephron</p> | <p>146. Germinal epithelium cells are present in :-<br/>         (1) Uriniferous tubules<br/>         (2) Oesophagus<br/>         (3) Inner lining of trachea<br/>         (4) Seminiferous tubules</p> <p>147. Microvilli of epithelium cells<br/>         (1) Push mucus<br/>         (2) Engulf the foreign matter<br/>         (3) Give movement to the cells<br/>         (4) Increase surface area</p> <p>148. The ducts of most of the glands and tubular parts of nephron in kidney are lined by :-<br/>         (1) Cuboidal epithelium<br/>         (2) Columnar epithelium<br/>         (3) Stratified squamous<br/>         (4) Ciliated squamous</p> <p>149. Which of the following is the main function of compound epithelium?<br/>         (1) Protection (2) Absorption<br/>         (3) Reproduction (4) Secretion</p> <p>150. Which statement is correct regarding compound epithelium?<br/>         (1) All the cells rest on the basement membrane<br/>         (2) Single layer of cells<br/>         (3) It consists of the similar type of cells<br/>         (4) It is mainly protective</p> <p>151. Choose the correct location of the lining epithelium which lacks basement membrane<br/>         (1) Lining of small intestine<br/>         (2) Lining of ureter<br/>         (3) Lining of buccal cavity<br/>         (4) Wall of blood vessel</p> <p>152. The cuboidal epithelium cells of ..... have microvilli which increase absorption surface area :-<br/>         (1) oviduct<br/>         (2) Proximal convoluted tubule<br/>         (3) Gall bladder<br/>         (4) Blood vessel</p> |
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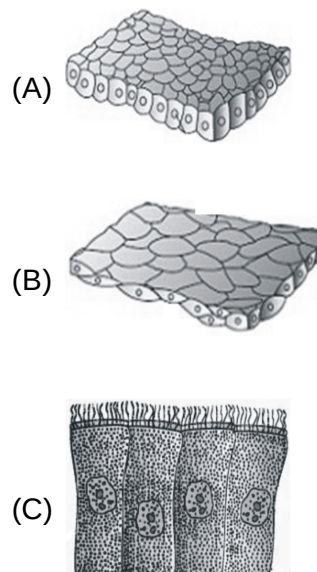
153. Following is the diagram of an epithelium, Choose the correct option w.r.t. the location where this type of epithelium is found :-



- (1) Bronchioles  
(2) Fallopian tubes  
(3) Ventricles of brain  
(4) All of these
154. Which of the following is incorrect?
- (1) Gap junction allow rapid transfer of ions  
(2) Adhering junctions are also called desmosomes  
(3) Tight junctions prevent leakage of material across the tissue  
(4) Gap junctions are also called zonula occludens
155. Stratified epithelium has many layer of cells, however the deepest layer is made up of columnar or cuboidal cells. Thus epithelium is classified on the basis of :-
- (1) Shape of the cells present in the middle layer  
(2) Total number of layers  
(3) Shape of the cells present in the superficial layer  
(4) Shape of the cells in the lower most layer
156. The term histology, i.e. study, of tissue was given by
- (1) Malpighi (2) Mayer  
(3) Harvey (4) Bichat
157. Which of the following statement is not applicable for pseudostratified epithelium?
- (1) It appears like multilayered because nuclei lie at different level  
(2) The long cells have oval nuclei however short cells have rounded nuclei

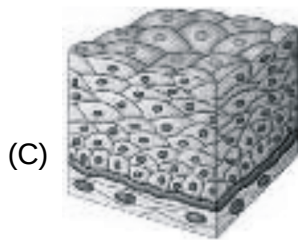
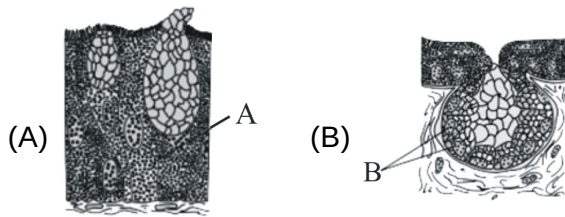
- (3) Short cells have cilia at their free surface, however the long cells are without cilia  
(4) It is found in trachea

158. If we scrape the lining of buccal cavity by sterilized forceps, what type of cells are is likely to be observed in the microscope?
- (1) Non-keratinized stratified squamous epithelium  
(2) Stratified columnar epithelium  
(3) Pseudostratified epithelium  
(4) Keratinized stratified squamous epithelium
159. Which of the following cell junction perform cementing to keep the neighbouring cells together?
- (1) Gap junction  
(2) Tight junction  
(3) Plasmodesmata  
(4) Adhering junction
160. Identify, A, B and C in the diagram given below :-



|     | A               | B               | C                        |
|-----|-----------------|-----------------|--------------------------|
| (1) | Simple columnar | Simple squamous | Simple ciliated          |
| (2) | Simple squamous | Simple columnar | Simple cuboid            |
| (3) | Simple cuboidal | Simple squamous | Simple ciliated columnar |
| (4) | Simple cuboidal | Simple columnar | Simple ciliated cuboidal |

161. Identify A, B and C in the given diagram and choose the correct combination of options :-



|     | A                   | B                   | C                              |
|-----|---------------------|---------------------|--------------------------------|
| (1) | Unicellular gland   | Multicellular gland | Transitional epithelium        |
| (2) | Multicellular gland | Unicellular gland   | Stratified squamous epithelium |
| (3) | Unicellular gland   | Multicellular gland | Stratified squamous epithelium |
| (4) | Multicellular gland | Unicellular gland   | Transitional epithelium        |

162. Which of the following statements is/are incorrect regarding digestion and absorption of food in human beings :-

- (a) About 90% of starch is hydrolysed by salivary amylase in our oral cavity
  - (b) Entero-endocrine cells in our stomach secrete the proenzyme trypsinogen
  - (c) Vitamin-D is produced in human body in skin
  - (d) Bile salts act as activator of pancreatic lipase
- (1) Two, a and b  
(2) Two, a and c  
(3) Two, a and d  
(4) Three, a, b and d

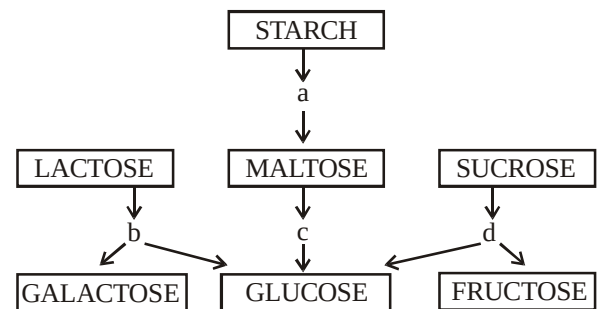
163. Which of the following reaction takes place in duodenum of small intestine :-

- (1) Dipeptides  $\rightarrow$  Amino acids
- (2) Maltose  $\rightarrow$  Glucose + Glucose
- (3) Triglycerides  $\rightarrow$  Monoglycerides + Fatty acid
- (4) Lactose  $\rightarrow$  Glucose + Glucose

164. Pepsin differs from trypsin in that it digest :-

- (1) Proteins in acidic medium in stomach
- (2) Proteins in alkaline medium in intestine
- (3) Proteins in acidic medium in intestine
- (4) Proteins in alkaline medium in stomach

165. Identify enzymes a, b, c and d in the digestion of carbohydrates :-



|   | a       | b       | c       | d         |
|---|---------|---------|---------|-----------|
| 1 | Lactase | Amylase | Lactase | Invertase |
| 2 | Lipase  | Lactase | Maltase | Invertase |
| 3 | Amylase | Lactase | Maltase | Sucrase   |
| 4 | Amylase | Lactase | Maltase | Lipase    |

166. Enterokinase is :-

- (1) nondigestive enzyme secreted by duodenal serosa
- (2) nondigestive enzyme secreted by duodenal mucosa
- (3) nondigestive enzyme secreted by duodenal muscularis layer
- (4) nondigestive enzyme secreted by mucosa of jejunum and ileum

167. Read the following statements :-

- (I) Absorption of water depends on osmotic gradient
- (II) Glucose and sodium ions are absorbed through active transportation
- (III) Fructose is absorbed by simple diffusion

Out of these which is/are not correct statements ?

- (1) Only I                                      (2) Only-II
- (3) Only-III                                      (4) Both I & III

168. Here given the list of various digestive enzymes:-

Pepsin, Trypsinogen, ptyline, chymotrypsinogen, procarboxypeptidase, nucleosidase, nuclease, amylase, dipeptidase and Lipase.

Out of these how many enzymes are found in pancreatic juice ?

- (1) Four                                      (2) Five
- (3) Six                                      (4) Seven

169. Read the following statements :-

- (a) Human liver is the largest endocrine gland of body and having two lobes
- (b) Hepatic lobules are structural and functional unit of liver and contained hepatocytes which are arranged in chord like manner
- (c) Glisson's capsule is covering of each lobules and made up by connective tissue
- (d) Bile juice is formed and secreted by hepatocytes and is stored into liver sinusoid

Out of these which statements are corrects and incorrects ?

- (1) Statements a and b are correct while c & d are incorrect
- (2) Statements a and d are correct while b & c are incorrect
- (3) Statements b & c are correct while a & d are incorrect
- (4) Statements b and d are correct while a & c are incorrect

170. Given below a table in which name of digestive enzymes, their substrate and their products are given in column I, II and III respectively. Select the option in which correctly matched pairs are given ?

| S. No. | Column-I<br>Name of enzyme | Column-II<br>Substrate | Column-III<br>Product        |
|--------|----------------------------|------------------------|------------------------------|
| (i)    | Maltase                    | Maltose                | Glucose + Fructose           |
| (ii)   | Lactase                    | Lactose                | Glucose + galactose          |
| (iii)  | Nucleosidase               | Nucleotide             | Sugar +N- Bases              |
| (iv)   | Sucrase                    | Sucrose                | Glucose + Fructose           |
| (v)    | Nucleotidase               | Nucleotide             | Nucleoside + phosphate group |

Options :-

- (1) (i), (ii), (iv) & (v)
- (2) (i), (ii), (iii) & (v)
- (3) (ii), (iv) & (v)
- (4) (i), (iii) & (iv)

171. Read the following statements :-

- (a) Chylomicrons are small lipoprotein particles that are absorbed from intestine into blood capillaries.
- (b) About 60% of starch is hydrolised by salivary amylase in our buccal cavity
- (c) Chief cells in our stomach secretes HCl and castle factor
- (d) Fructose and some amino acids are absorbed through intestinal mucosa by the help of carrier proteins

Select the option which includes only correct statement/s ?

Options :-

- (1) Statements (a) & (b)
- (2) Statements (c) & (d)
- (3) Statement (a) only
- (4) Statements (d) only

172. Which one of the following is the correct match of the site of the action on the given substrate, the enzyme acting upon it and the end product?

- (1) Duodenum :→ Triglycerides  $\xrightarrow{\text{Trypsin}}$  Monoglycerides  
 (2) Small intestine :→ protein  $\xrightarrow{\text{Pepsin}}$  Amino acid  
 (3) Stomach :→ Nucleic acid  $\xrightarrow{\text{Nuclease}}$  nucleotide  
 (4) Ileum :→ Dipeptides  $\xrightarrow{\text{Dipeptidase}}$  Amino acid

173. Consider the following four statement (a-d) and select the option which includes all the correct ones only.

- (a) At the time of need, liver cells convert the non carbohydrates compound into glycogen is called glyconeogenesis  
 (b) When glucose level in blood falls down then liver cell convert glycogen into glucose is known as glycogenolysis  
 (c) Extra amount of glucose is converted into glycogen by liver cells is known as glycogenesis  
 (d) Synthesis of glycogen from lactic acid by liver cell is known as glyconeogenesis

Option -

- (1) Statement (a), (b), (c) and (d)  
 (2) Statement (b), (c) and (d)  
 (3) Statements (a), (c) and (d)  
 (4) Statements (a) and (d)

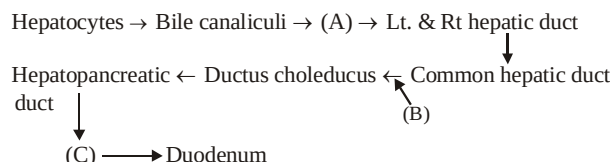
174. Read the following statements :-

- (i) "Sphinctre of oddi" controls & regulates the secretion of only pancreatic juice  
 (ii) Sphinctre of boyden" controls & regulates the secretion of only bile juice  
 (iii) "Sphinctre of oddi" present on ampulla of vater which controls and regulates the secretion of bile juice as well as pancreatic juice  
 (iv) "Sphinctre of Boyden" present on common hepatic duct

On the basis of above statements select the correct option which includes all the correct ones ?

- (1) Only (i) & (ii)  
 (2) Only (ii) & (iii)  
 (3) Only (iii) & (iv)  
 (4) Only (i) & (iv)

175. Given below the flow chart of pathway of bile juice in human being :-



On the basis of flow chart, select the option in which alphabets A, B and C are correctly identified according to their name ?

- (1) A = branch of hepatic duct  
 B = Common bile duct  
 C = Ampulla of vater  
 (2) A = Branch of hepatic duct  
 B = Main pancreatic duct  
 C = Ampulla of vater  
 (3) A = Cystic duct  
 B = Branch of hepatic duct  
 C = Main pancreatic duct  
 (4) A = Branch of hepatic duct  
 B = Cystic duct  
 C = Ampulla of vater

176. Given, 1 = natural selection, 2 = variations and their inheritance, 3 = survival of the fittest, 4 = struggle for existence. According to Darwinism, which of the following represents the correct sequence of events in the origin of a new species?

- (1) 1, 2, 3, 4                      (2) 2, 3, 1, 4  
 (3) 3, 4, 1, 2                      (4) 4, 2, 3, 1

177. Which one proposed that the first from of life came from pre-existing non-living orgnic molecules ?

- (1) Oparin and Haldane  
 (2) Stanely Miller and Herald Urey  
 (3) Lamarek and Darwin  
 (4) Hugo de Vries and Dobzhansky

178. The process of evolution of different species in a given geographical area starting from a point and literal radiating to other areas of geography (habitats) is called

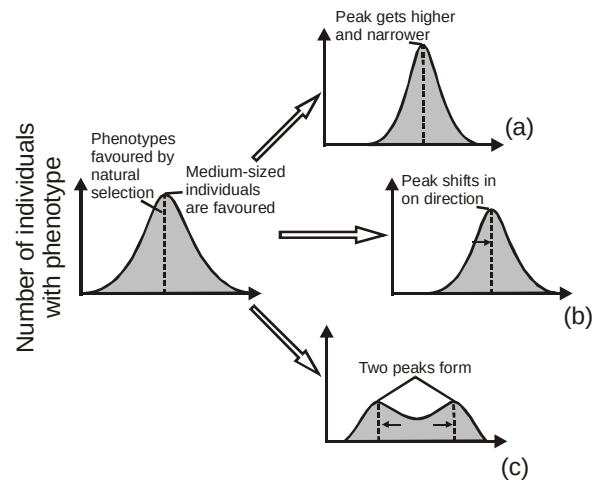
- (1) Adaptive convergence
- (2) Adaptive radiation
- (3) Natural selection
- (4) Convergent evolution

179. Fill in the blanks according to the convergent evolution

| Placental mammals | Australian mammals  |
|-------------------|---------------------|
| Anteater          | ....a....           |
| ....b....         | Spotted cuscus      |
| Flying squirrel   | ....c....           |
| .....d....        | Tasmanian tiger cat |

- (1) c-flying phalanger, b-lemur, d-bobcat, a-Numbat
- (2) b-flying phalanger, a-lemur, c-bobcat, d-Numbat
- (3) c-flying phalanger, a-lemur, d-bobcat, b-Numbat
- (4) d-flying phalanger, b-lemur, c-bobcat, a-Numbat

180. Recognise the figure and find out the correct matching.



- (1) b-directional, a-disruption, c-stabilisation
- (2) b-directional, c-disruption, a-stabilisation
- (3) c-directional, b-disruption, a-stabilisation
- (4) a-directional, c-disruption, b-stabilisation